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November 1, 1962

INTERIM FEDERAL SPECIFICATION

COUNTER, RECIPROCATING, HAND HELD

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand held, mechanical, reciprocating counters generally used for counting people, vehicles, trees, packages, blood cells, bateriological colonies, and other visible objects.

1.1.1 Federal specification coverage. Federal specifications do not cover all types of the commodity indicated by the titles of the specification, but are intended to cover the types ordinarily used by the Federal Government.

1.2 Classification. Counters covered by this specification shall be four digit ("0000" to "9999" registering range), and of the following types, as specified (see 3.6 and 6.1):

Type I - High Count (capable of functioning for at least 5,000,000 counts without failure).

Type II - General Purpose (capable of functioning for at least 500,000 counts without failure).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes; Wood, Wirebound.
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-P-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed, and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes: Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Setup.
PPP-T-76 - Tape, Pressure Sensitive Adhesive Paper, (For Carton Sealing).

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

FSC 6680

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling Porcedures and Tables For Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Qualification. Counters furnished under this specification shall be a product which has passed the qualification inspections and tests specified in 4.3, and has been listed or approved for listing on the applicable Qualified Products List.

3.2 Description. The counter shall be of a wheel register type, designed to be held in one hand and actuated by the thumb of same hand. The force required to actuate counter (depress counter operating key and advance the counter register wheels) shall not exceed 3-pounds when determined in accordance with 4.6.3.

3.3 Material. Material shall be as specified herein. Material not definitely specified shall be of a good commercial quality, entirely suitable for the purpose intended. Material shall be free from all defects and imperfections that might affect the serviceability of the finisned product (see 4.3.2 and 4.5).

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3.4 Marking. The counter shall be permanently marked with the manufacturer's name or trademark. Marking characters shall be of such size as to permit reading with naked-eye (see 4.3.2 and 4.5).

3.5 Construction. Essentially, the counter shall consist of a case, registering wheels and count mechanism, operating key, reset mechanism, and a means of holding securely, i.e., finger ring, hand or wrist strap or chain. If the holding means is a finger ring, the ring shall be attached to the case and shall swing out to permit finger insertion when counter is operated by either hand. After advancement or reset, the four digit numeral appearing in counter window shall be symmetrically located within the window periphery and the alignment of any individual numeral with respect to any other numeral shall vary by no more than 1/32 inch (see 4.3.2 and 4.5). When specified (see 6.1), each counter shall be provided with a portable support stand designed to hold the counter in correct operating position, precluding the necessity for holding counter in hand. The stand shall be such as to support counter on a horizontal work surface.

3.5.1 Case. The case shall be of corrosion-resistant material and shall not show evidence of breaking, chipping, or cracking when tested, as specified in 4.6.1. The case shall have a viewing window made of clear glass or clear plastic through which the register numerals can be easily read. Additionally, case shall be designed to fit comfortably in either hand in such manner as to permit ease of operation. Also, case shall be designed to preclude internal collection of dust and grime, with openings where necessary and only large enough to permit counter to function properly (see 4.3.2 and 4.5).

3.5.2 Registering wheels. Each registering wheel shall be provided with ten numerals from 0 to 9 inclusive. The numerals shall be not less than 5/32 inch in height, permanently marked, well-proportioned, and for legibility, of a color contrasting to the background. Means shall be provided whereby the units, tens, hundreds, and thousands digits shall be advanced in proper sequence when the wheels are operated by the operating key (see 4.3.2 and 4.5).

3.5.3 Operating key. The operating key, for advancing numerals, shall be designed to permit ease of operation by thumb of either hand and shall be either a plunger or lever projecting outside the case. Each depression of the key shall cause the counter register to advance one consecutive numeral without skipping any numeral regardless of the position in which counter is operated (see 4.3.2, 4.5 and 4.6.2).

3.5.4 Holding ring. If supplied with a holding ring, the inside diameter must be 1 $\pm$ 3/16 inch and the material shall be at least 0.06-inches thick (see 4.5).

3.6 Count life. A type I counter shall be capable of functioning for at least 5,000,000 counts without failure when tested, as specified in 4.6.4. A type II counter shall be capable of functioning for at least 500,000 counts without failure when tested, as specified in 4.6.4. Classification of counters shall be determined at the time of qualification inspections and tests (see 3.1 and 4.3).

3.7 Reset function. Counters shall be provided with a means whereby all registering wheels can be quickly and accurately returned to a zero reading. The resetting of all registering wheels to zero reading shall be accomplished by not more than two revolutions of a reset knob or two depressions of a reset button. Zero reading shall be a reading of "0000" in the counter window. If a reset knob is used, it shall have a knurled or serrated periphery for ease of turning (see 4.5).

3.8 Finish. Unless otherwise specified (see 6.1), the counter shall be provided with a corrosion-resistant finish (see 4.5).

3.9 Workmanship. Counters shall conform to the quality and grade of the product established by this specification. All burrs and rough and sharp edges shall be removed. In general, counters shall be free from any defects that affect appearance or serviceability or any defects that may cause injury to the user (see 4.3.2 and 4.5).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections and tests. The inspection and testing of counters covered by this specification shall be classified as follows:

A. Qualification inspection and testing (see 4.3).
B. Lot acceptance inspection and testing (see 4.4).

4.3 Qualification inspection and testing.

4.3.1 Sampling. Thirty three (33) counters of the same type, design, and manufacture shall be randomly selected from the manufacturer's current production and shall be subjected to the following inspections and tests in sequential order, as numbered:

(1) Subjective inspection in accordance with 4.3.2.
(2) Visual inspection as indicated in 4.3.3.
(3) Performance testing as specified in 4.3.4.

4.3.2 Subjective inspection. Ten of the samples, selected in accordance with 4.3.1, shall be subjectively inspected to determine compliance with the requirements cited below. Failure of any counter to meet any requirement shall preclude qualification, whereupon further inspection or testing shall not be performed. The requirements for subjective inspection are:

(a) The counter shall be designed with a means for holding securely.

(b) The case shall have a viewing window or windows through which the register numerals can be easily read.

(c) Case shall be designed to fit comfortably in either hand in such manner as to permit ease of single hand operation.

(d) Case shall be designed to preclude dust and grime, with openings where necessary and only large enough to permit counter to function properly.

(e) The numerals shall be permanently marked, well proportioned, and for legibility, of a color contrasting to background.

(f) The operating key, for advancing numerals, shall be designed to permit ease of operation by the thumb of either hand.

(g) Counters shall be free from any defects that affect appearance or serviceability or any defects that may cause injury to the user.

4.3.3 Visual inspection. Thirteen (13) of the samples selected in accordance with 4.3.1 shall be subjected to the visual inspection test specified in 4.5. Any counter with one or more major or minor defects shall be considered defective. If the sample contains any counter(s) with a major defect(s) or more than one counter with minor defect(s), qualification shall be precluded. Subsequent to visual inspection, counters shall be subjected to tests as specified in 4.3.4 regardless of whether qualification is precluded at time of visual inspection.

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4.3.4 Tests. All 33 of the samples selected in accordance with 4.3.1 shall be subjected to the performance tests specified in 4.6, 13 of the samples being subjected to all tests specified in 4.6.1, 4.6.2 and 4.6.3 and 20 of the samples being subjected only to the test specified in 4.6.4. Failure of more than one counter to pass any of the tests specified in 4.6.1, 4.6.2 or 4.6.3 shall preclude qualification. For testing as specified in 4.6.4, qualification criteria shall be dependent upon counter type (see 1.2). The distinction between types shall be made on the basis of capability to function for 5,000,000 counts (type I) or 500,000 counts (type II). Therefore, for the 20 samples subjected to the 4.6.4 test, qualification criteria shall be as follows:

(1) For type I, not more than two failures under 5,000,000 counts shall be permitted, provided that no more than one of the two failures occurs under 1,000,000 counts.

(2) For type II, not more than two failures under 500,000 counts shall be permitted, provided that no more than one of the two failures occurs under 100,000 counts.

4.3.5 Disqualification. The Government reserves the right to remove from the Qualified Products List (QPL), any counters that have passed qualification inspections and tests and subsequently fail to pass lot acceptance inspection and testing specified in 4.4.

4.3.6 Test location and expenses. Qualification inspections and tests shall be conducted at a Government laboratory designated by the Standardization Division, Federal Supply Service, General Services Administration. Qualification inspection and test expenses shall be borne by manufacturer or supplier.

4.4 Lot acceptance inspection and testing. Lot acceptance inspection and testing shall consist of a visual inspection test and performance tests as delineated in 4.5 and 4.6, respectively.

4.4.1 Inspection and test lot. The inspection and test lot shall consist of all units of product of the same type, design and manufacture submitted for inspection and testing at one time.

4.4.2 Sampling and acceptable quality levels. Sampling for lot acceptance inspection and tests shall be performed in accordance with MIL-STD-105. Visual inspection (4.5) shall be performed using inspection level S-3 and an acceptable quality level (AQL), expressed in percent defective, of 1.0 for Major Defects and 4.0 for Minor Defects. For tests specified in 4.6.1, 4.6.2, and 4.6.3 samples shall be selected in accordance with inspection level S-3 and subsequently subjected to all three tests; i.e., the samples selected for testing in accordance with 4.6.1 shall also be subjected to 4.6.2 and 4.6.3 tests. The acceptable quality level (AQL) for 4.6.1, 4.6.2 and 4.6.3 tests shall be 4.0, expressed in percent defective. For the test specified in 4.6.4, the inspection level shall also be level S-3 and the acceptable quality level (AQL) shall be 2.5, expressed in percent defective; however, the samples selected shall only be subjected to the 4.6.4 test; i.e., test samples selected for 4.6.4 testing shall be separate from those selected for testing as specified in 4.6.1, 4.6.2 and 4.6.3.

4.4.3 Count life test duration. For lot acceptance, each type I counter shall be tested, as specified in 4.6.4, for 1,000,000 counts or until failure occurs; and each type II counter shall be tested for 100,000 counts or until failure occurs.

4.5 Visual inspection test. Visual inspection shall be conducted in accordance with table I. Major and Minor defects shall be as indicated in the table.

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TABLE I. Classification of defects

Examine	Defect	Major	Minor
Case	Not made of corrosion-resisting material...	X	
	Manufacturer's name or trademark not legible or permanently marked.....		X
	Viewing window not as specified.....	X	
Registering	Not provided with ten numerals from 0 to 9.....	X	
	Numerals less than 5/32 inch in height....	X	
	Not in contrasting color and clearly legible.....	X	
	Numerals not permanently marked.....	X	
	Four digit numerals not symmetrically located within window.....	X	
	Individual numerals out of alignment with respect to each other by more than 1/32 inch.....	X	
	Tens, hundreds, or thousands digits do not advance when preceding digit moves from 9 to 0.....	X	
Operating key	Not specified type of key (see 3.5.3).....	X	
	Binds, sticks, or does not operate with ease.....	X	
Holding ring (if used)	Ring material less than 0.06 inches in thickness.....		X
	Inside diameter of ring less than 13/16 or more than 1-3/16.....		X
Counter	Component loosely attached.....		X
	Component missing.....	X	
	Component damaged, affecting serviceability.....	X	
	Component damaged, affecting appearance, but not affecting serviceability.....		X
	Burrs and sharp edges.....	X	
Reset function	Not provided with reset to zero, as specified (see 3.7).....	X	
	Reset knobs not knurled or serrated.....		X
Workmanship	Not as specified (see 3.8).....	X	

4.6 Performance testing. Tests specified in 4.6.1, 4.6.2 and 4.6.3 shall be performed in sequence as numbered. Any counter failing any of the below tests shall be considered defective.

4.6.1 Drop test. The completely assembled counter shall be dropped five times from a height of three feet onto a concrete floor and shall then be examined for evidence of breaking, chipping, or cracking. Each drop is to be free fall and shall be such as to permit the counter to strike the floor in a random manner. Dents in the case will be permitted as long as the counter passes the accuracy test (4.6.2), and actuating force test (4.6.3). Evidence of breaking, chipping, or cracking shall constitute a failure and the counter shall be considered defective.

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4.6.2 Accuracy test. After testing in accordance with 4.6.1, the counter shall be set to the zero reading, i.e., "0000", whereupon the inspector shall cover the viewing window and actuate the counter's operating key 100 times. Actuation shall be accomplished at a rate of approximately 100 counts per minute. After 100 actuations the inspector shall uncover the viewing window and verify whether or not the counter registers 100 counts. A count reading other than 100 shall constitute a failure and the counter shall be considered defective. The inspector may repeat the above operation several times to insure that the counter functions in the manner prescribed.

4.6.3 Determination of actuating force. Position counter register wheels for a window reading of "0999". Advance the counter register wheels, by depressing counter operating key, to a window reading of "1000" while simultaneously recording the force required to depress operating key. Any counter requiring application or more than 3 pounds force shall be considered defective.

4.6.4 Count life test. The count life test shall be conducted as follows:

(1) Place counter in test machine holder and position counter such that counter operating key will upon actuation by test machine cause counter to register a count increase of one each time key is depressed. Lock counter in this position and insure that body of counter (case) remains stationary during test.

(2) Adjust test machine actuating device such that a force of 3 pounds will be applied to counter operating key per each count actuation. Additionally, adjust test machine, such that, at the bottom of the actuating stroke, a clearance of approximately 0.015 inch exists between the bottom face of the actuating device and top of counter operating key when key is fully depressed or in its "bottomed out" position. Also, adjust test machine such that a clearance of at least 0.015 inch exists between top of counter operating key and bottom face of actuating device prior to each count actuation (depression of counter key).

(3) Adjust test machine to provide a cyclic rate of 200 $\pm$ 10 count actuations (depressions) of the counter operating key per minute.

(4) Position test machine actuating device so that counter operating key is free of actuating device; i.e., not depressed, and adjust counter for zero reading; i.e., a window reading of "0000". Also, set test machine master counter for zero reading.

(5) Place test machine in operation and begin testing.

(6) During test, testing shall be interrupted at the rate(s) indicated below, and the count on each counter under test shall be recorded and compared with the count indicated on the master counter. If at this time the count on any test counter(s) is not in agreement, within $\pm$ 2 counts, with the count indicated on the test machine counter, such counter shall be considered as having failed this test and shall be deemed defective. The rate of the test interruption shall be as follows:

(a) At the end of the first 100,000 counts.

(b) At the end of each successive 250,000 counts to completion of test.

(7) After each test interruption, the master counter and counters under test shall be reset to zero reading, whereupon, testing shall be resumed.

(8) The total counts recorded for a counter under test shall be the cumulative sum of the counts recorded for the counter after each test interruption (Step 6).

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4.7 Inspection of preparation for delivery. An inspection shall be made to determine whether packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table II. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection lot shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Any component missing or damaged.
Workmanship	Inadequate application of components such as closure of container flaps, loose strapping, inadequate stapling, or distortion of container.
Contents	Number per container is more or less than required; gross weight exceeds box specified requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Preservation and unit packaging. The counters shall be individually preserved and unit packaged in conformance with MIL-P-116, method 1A 3, without preservative, using boxes conforming to PPP-B-566, as the containers. The dip-coated sealed package shall be over-wrapped with kraft paper suitably secured.

5.1.1.2 Intermediate packaging. Fifty (50) counters, unit packaged, as specified in 5.1.1.1 shall be further packaged in a box conforming to PPP-B-636, class weather resistant, grade W6s and W6c. Suitable cushioning shall be provided to prevent movement within the boxes. The boxes shall be closed and sealed in conformance with the appendix to the box specification, and in addition, all joints and seams, including those occuring in the manufacture of the boxes, shall be covered with minimum 3-inch wide tape conforming to PPP-T-76, extended over all corners and edges at least 3 inches.

5.1.2 Level B.

5.1.2.1 Unit packaging. Each counter shall be packaged in a close fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. Each box shall be secured to prevent accidental opening.

5.1.2.2 Intermediate packaging. Ten (10) counters, packaged as specified in 5.1.2.1, shall be further packaged in a close fitting box conforming to PPP-B-566, PPP-B-665, PPP-B-676, or PPP-B-636. Each box shall be secured to prevent accidental opening.

5.1.3 Level C. Counters shall be packaged in accordance with the manufacturers standard practice, providing that this will insure protection for the items during shipment and provide for safe delivery to its destination, while complying with the common carriers rules as a minimum.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

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5.2.1 Level A. Counters, packaged as specified in 5.1, shall be packed together in quantities not to exceed 70 pounds gross weight in a close fitting box, conforming to PPP-B-621, style 2, 2-1/2, 3 or 4 or PPP-B-601, overseas type, style A or B for type 2 load. Suitable cushioning shall be provided to prevent movement within the boxes. The boxes shall be closed and strapped in conformance with the appendix to the applicable box specification.

5.2.2 Level B. Ten (10) intermediate packages (100 counters), packaged as specified in 5.1, shall be packed in a close fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with method II of the appendix to PPP-B-636.

5.2.3 Level C. The counters, packaged as specified in 5.1, shall be packed to assure carrier acceptance and safe arrival at destination in compliance with the National Motor Freight Classification rules or the Uniform Freight Classification rules, as applicable.

5.3 Marking. All marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.1).

5.3.1 Civil agencies. Interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Portable support stand, if required (see 3.5).
(d) Finish, if different (see 3.8).
(e) Inspection responsibility, if other than specified (see 4.1).
(f) Level of preservation, packaging and packing required (see section 5).
(g) Marking required (see 5.3).

6.2 At the option of the Contracting Officer, type I counters may be bid in competition with type II counters.

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